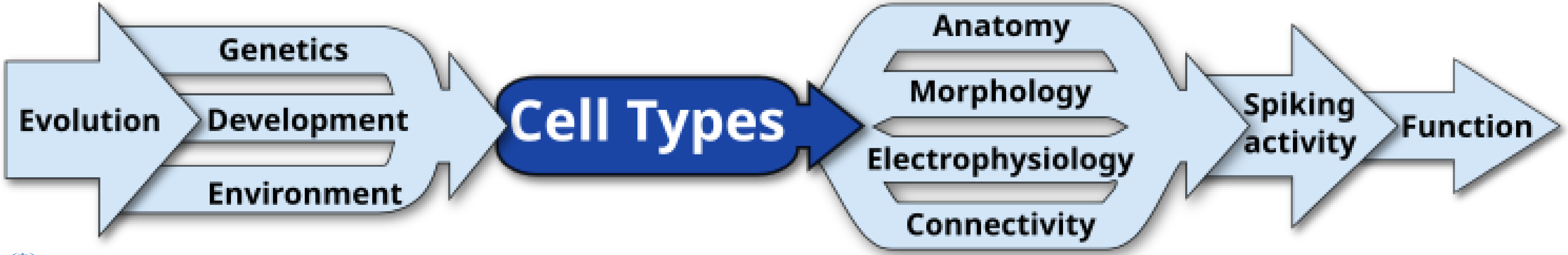
Cell Types Introduction

Describe Your Neurons Workshop, April 6th 2026

Rachel Hostetler, Scientist I

What are cell types?

- Cell types are **groups of cells with more similar quantitative cell properties to each other than to other cells**
- Cells in the same cell type **share features** defined on one or more measurable criteria
- However, we currently lack **standard definitions** of cell types in the brain



How were cell types historically defined?

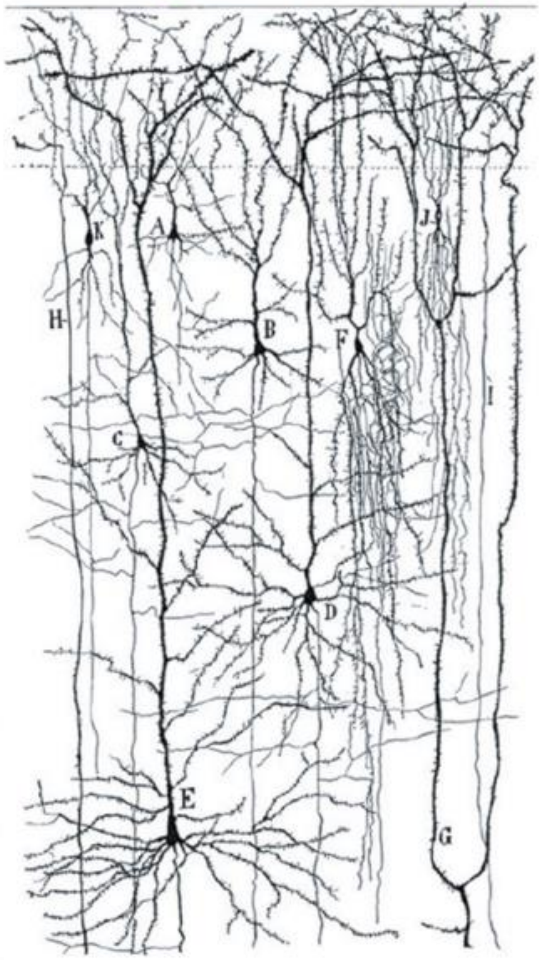
By morphology

Staining/visualizing cells and tracing their shapes

Cells were categorized and defined by their <u>shapes</u>

Examples:
Pyramidal,
Chandelier,
Martinotti,
Basket,
Bushy, etc.

Stereotypy of neuron morphology



Ramón y Cajal (1904), Yuste et al (2020)

How were cell types historically defined?

By connectivity

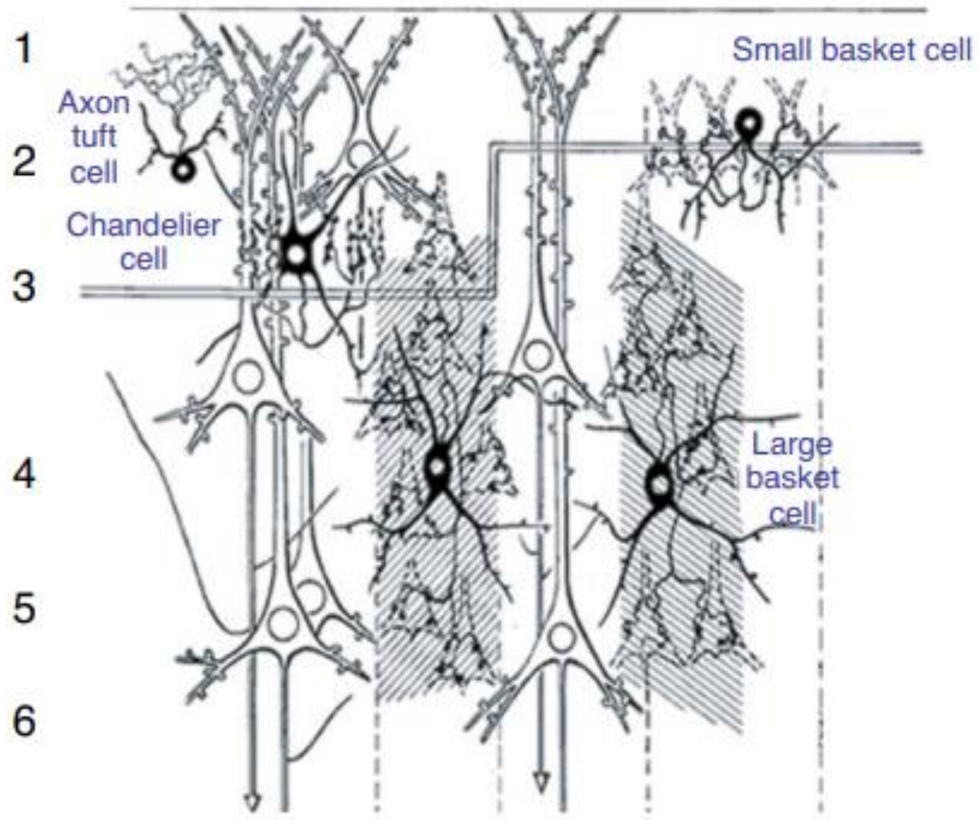
Staining/visualizing cells and tracing their connections

Cells were categorized and defined by their <u>connections to other cells</u>

Examples:
chandelier cells target axon initial segments

Basket cells target cell bodies

Stererotyped connectivity motifs



Szentágothai (1975), Yuste et al (2020)

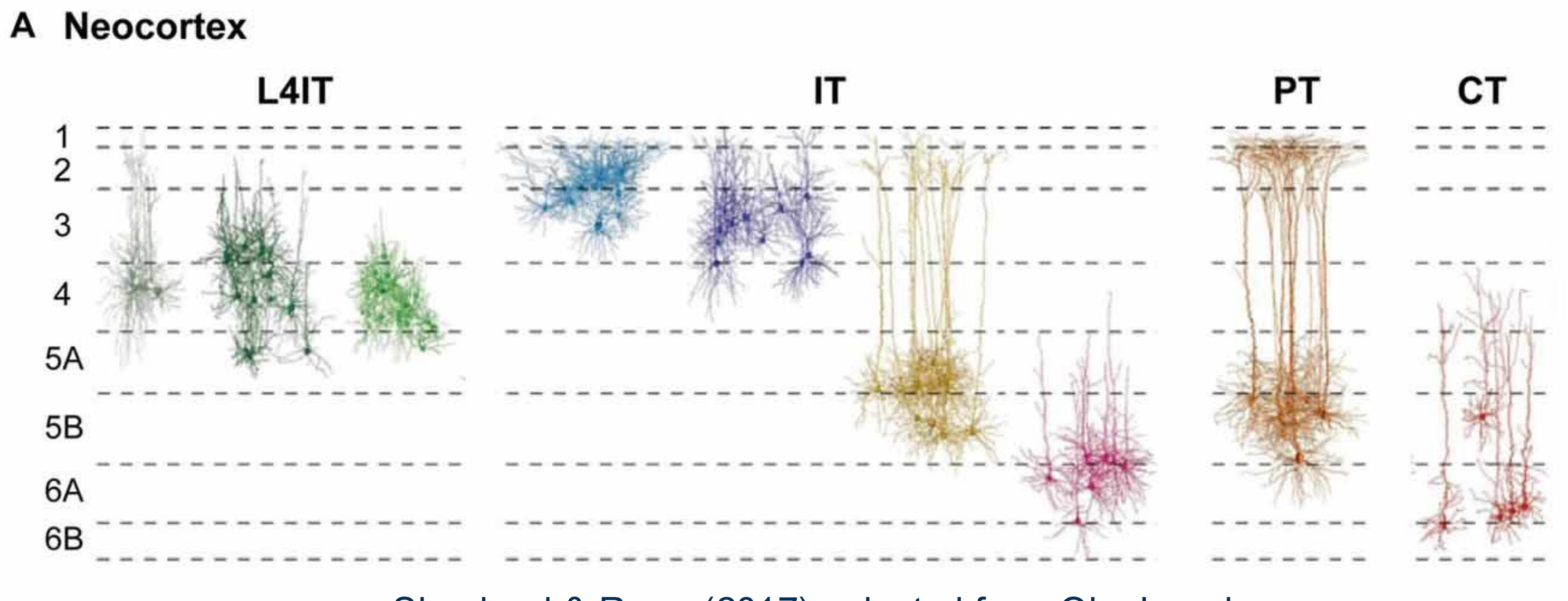
How were cell types historically defined?

By long-range projections

Using dyes for anterograde/retrograde tracing

Cells were categorized and defined by the <u>areas of the brain they project to</u>

Examples:
intratelencephalic (IT),
pyramidal tract (PT),
corticothalamic (CT)



Shepherd & Rowe (2017), adapted from Oberlaender et al (2012) and Harris & Shepherd (2015)

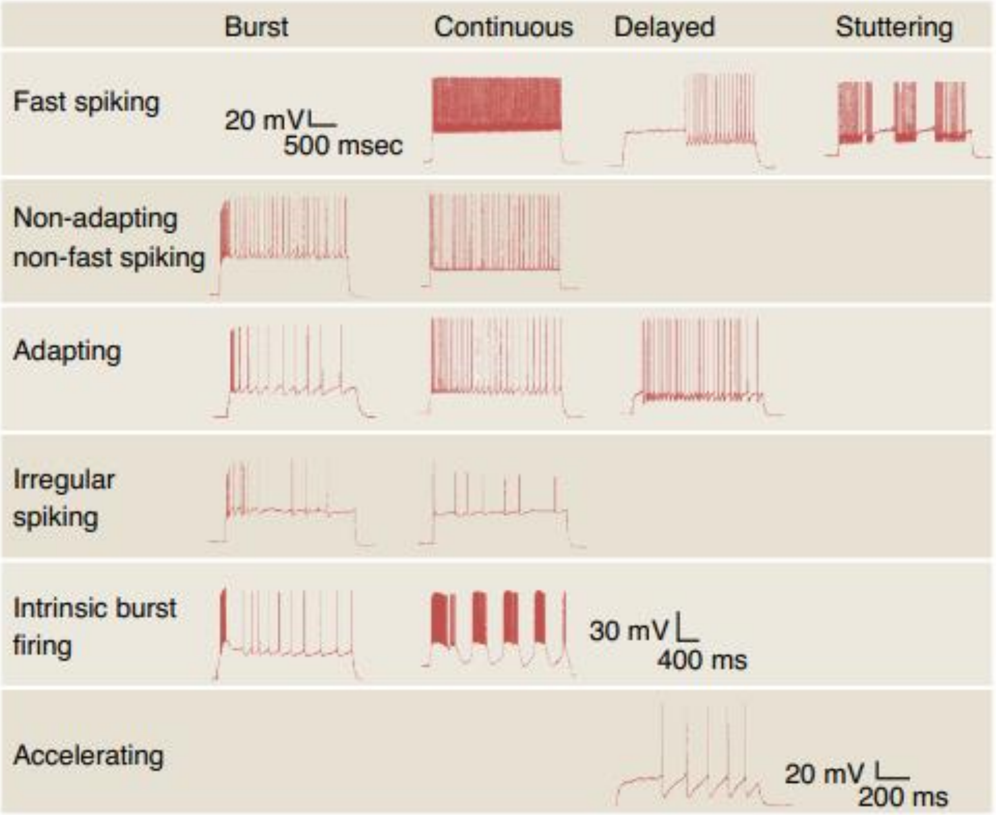
How were cell types historically defined?

By electrophysiology

Using patch-clamp to depolarize cells and record action potentials

Cells were categorized and defined by their <u>firing patterns</u>

Examples:
Fast spiking,
Adapting,
Bursting



Ascoli et al (2008)

How were cell types historically defined?

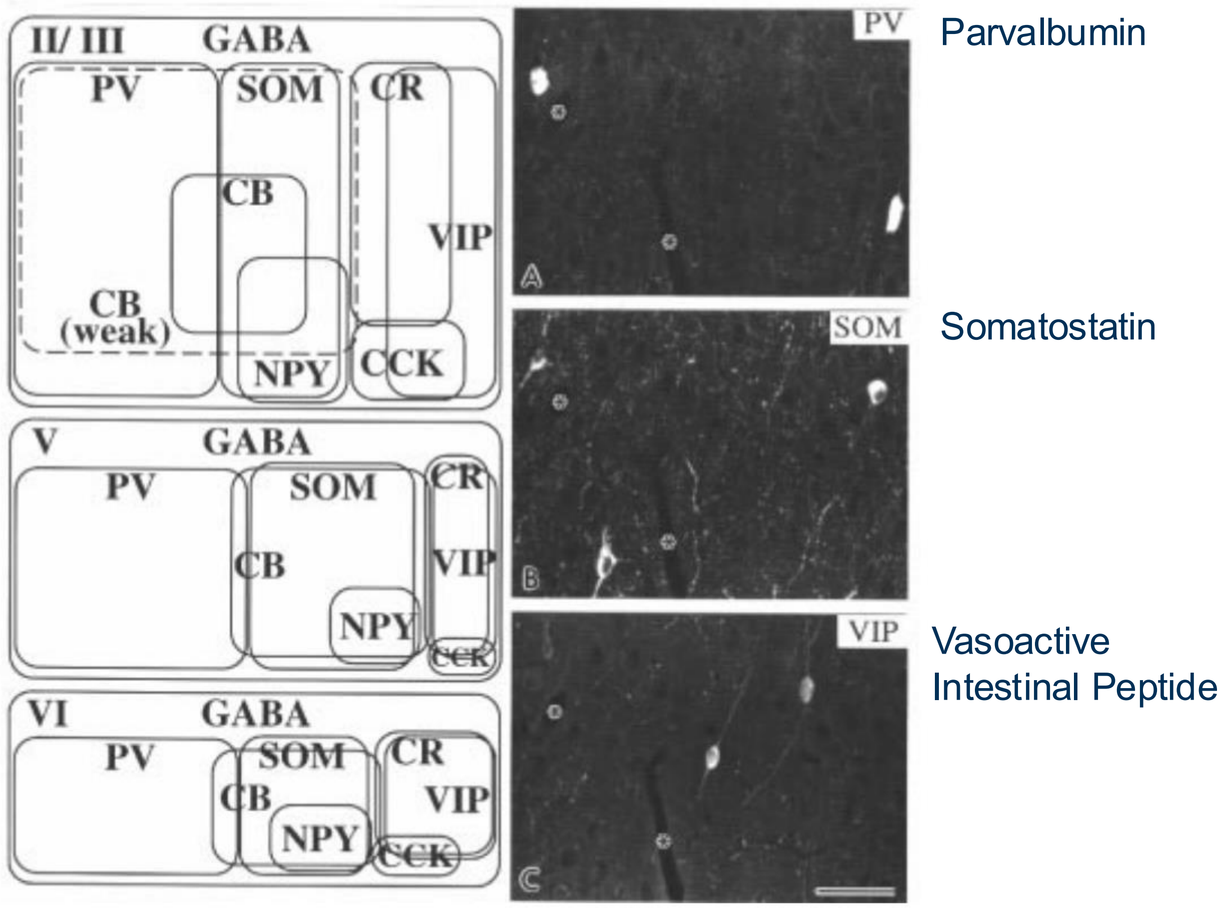
By molecular markers

Using immunohistochemistry to label proteins

Cells were categorized and defined by the <u>expression of molecular markers</u>

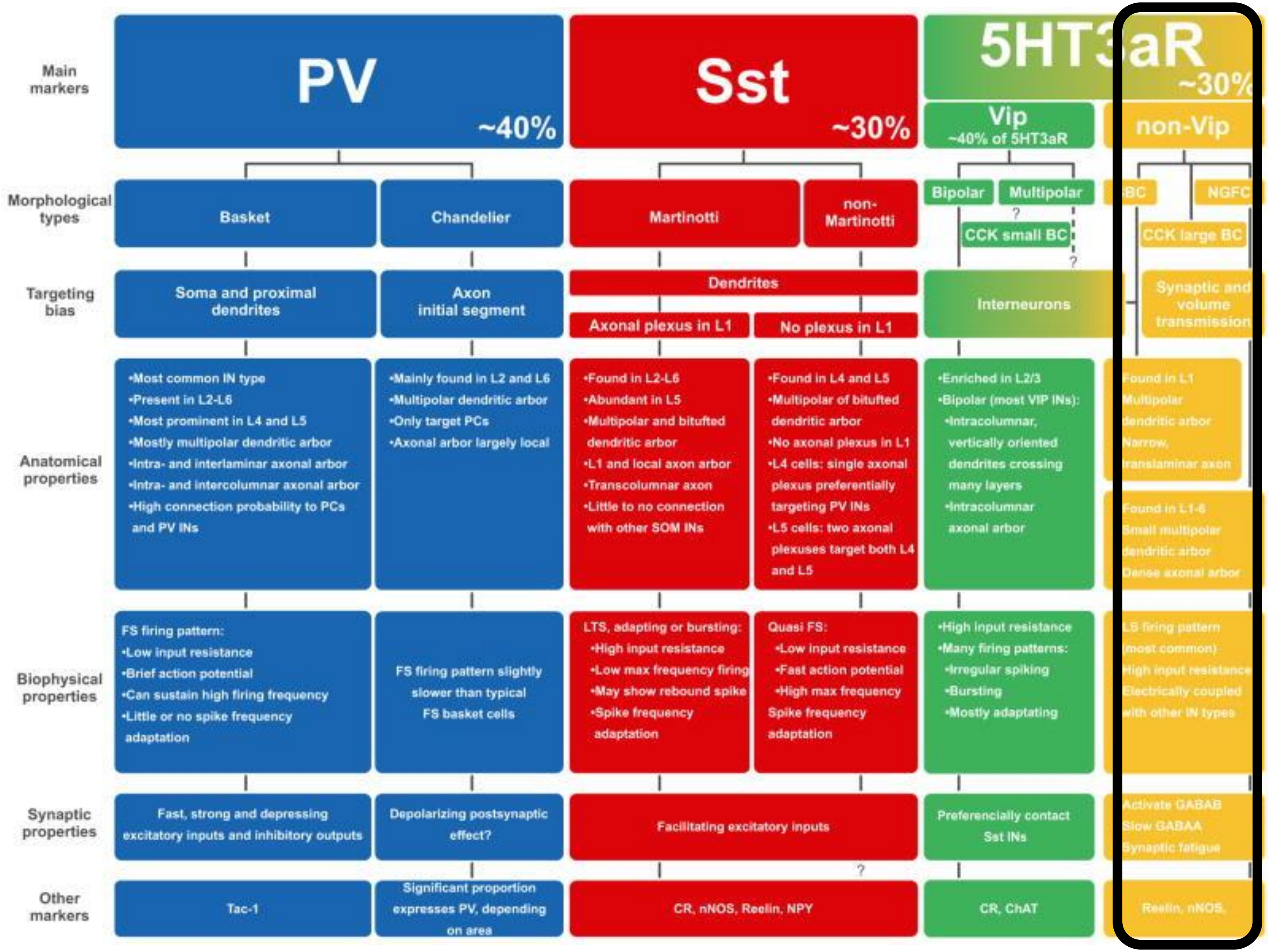
Examples:
SOM/SST cells express the neuropeptide somatostatin

PV cells express the calcium binding protein parvalbumin



Kawaguchi & Kubota (1997)

Combining historical modalities

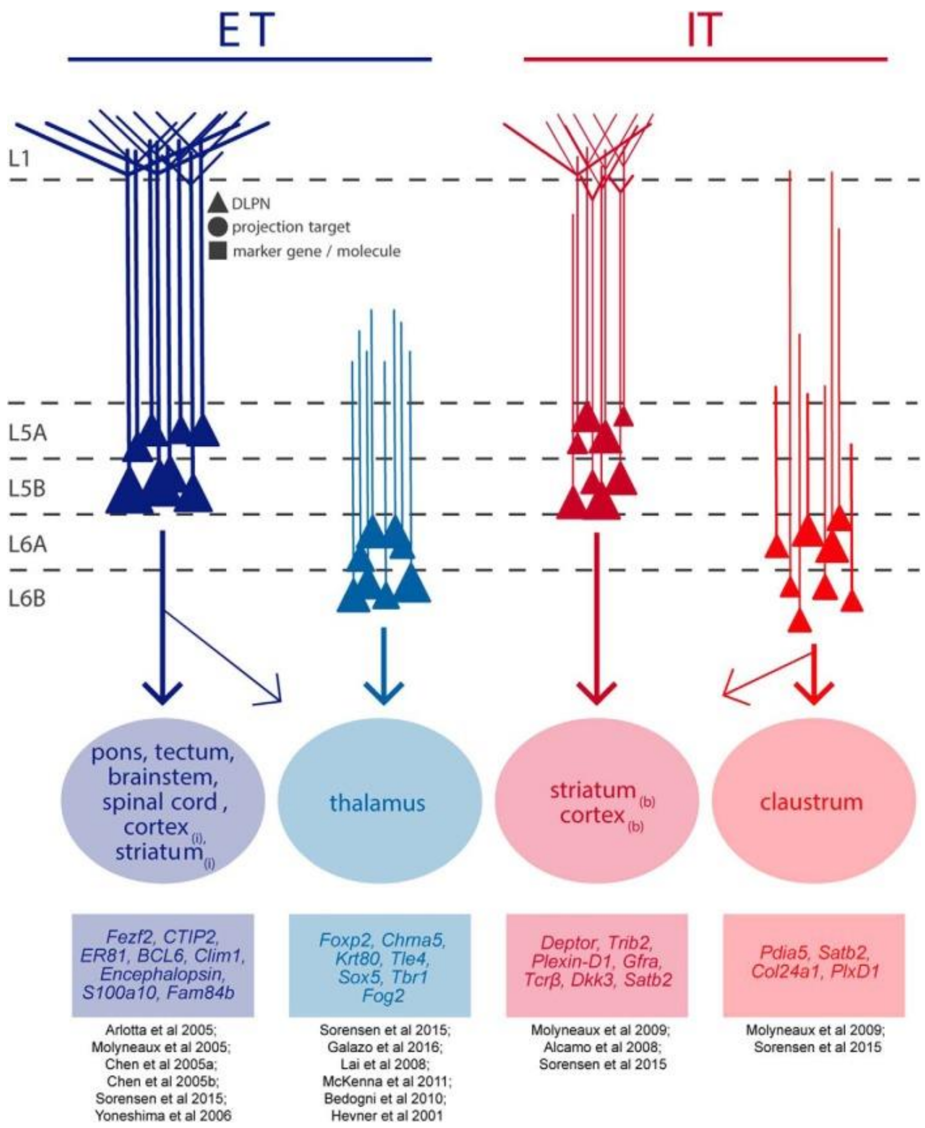


Id2 cells

Machold et al (2023)

Tremblay et al (2016)

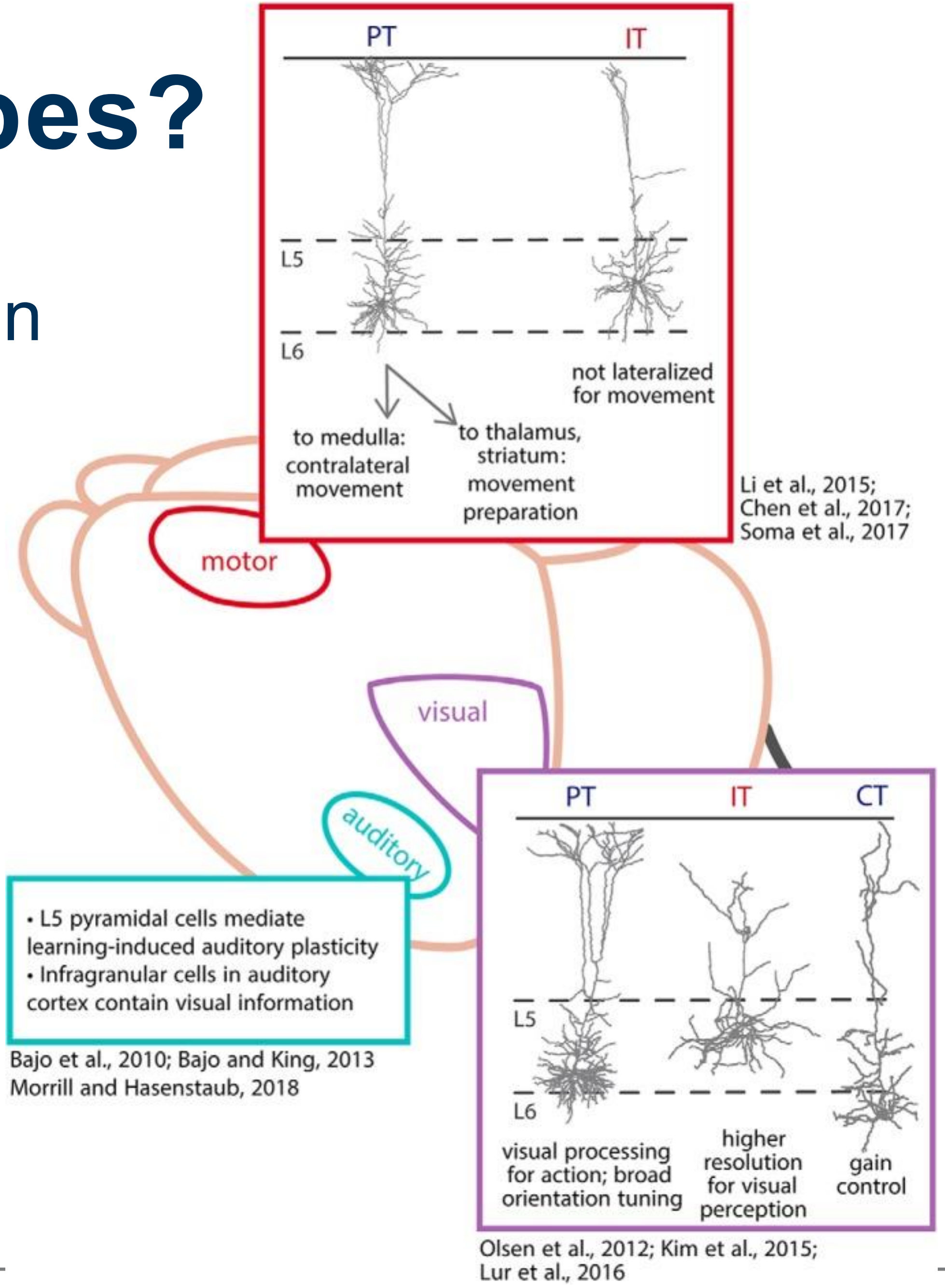
Combining historical modalities



Baker et al (2018)

Why study cell types?

They have distinct functions in motor planning and sensory processing



Baker et al (2018)

How to determine cell types?

Guidelines | Published: July 2008

Petilla terminology: nomenclature of features of GABAergic interneurons of the cerebral cortex

The Petilla Interneuron Nomenclature Group (PING)

Nature Reviews Neuroscience **9**, 557–568 (2008) | Cite this article

PING consists of: Giorgio A. Ascoli, Lidia Alonso-Nanclares, Stewart A. Anderson, German Barrionuevo, Ruth Benavides-Piccione, Andreas Burkhalter, György Buzsáki, Bruno Cauli, Javier DeFelipe, Alfonso Fairén, Dirk Feldmeyer, Gord Fishell, Yves Fregnac, Tamas F. Freund, Daniel Gardner, Esther P. Gardner, Jesse H. Goldberg, Moritz Helmstaedter, Shaul Hestrin, Fuyuki Karube, Zoltán F. Kisvárday, Bertrand Lambolez, David A. Lewis, Oscar Marin, Henry Markram, Alberto Muñoz, Adam Packer, Carl C. H. Petersen, Kathleen S. Rockland, Jean Rossier, Bernardo Rudy, Peter Somogyi, Jochen F. Staiger, Gabor Tamas, Alex M. Thomson, Maria Toledo-Rodriguez, Yun Wang, David C. West and Rafael Yuste.

How to determine cell types?

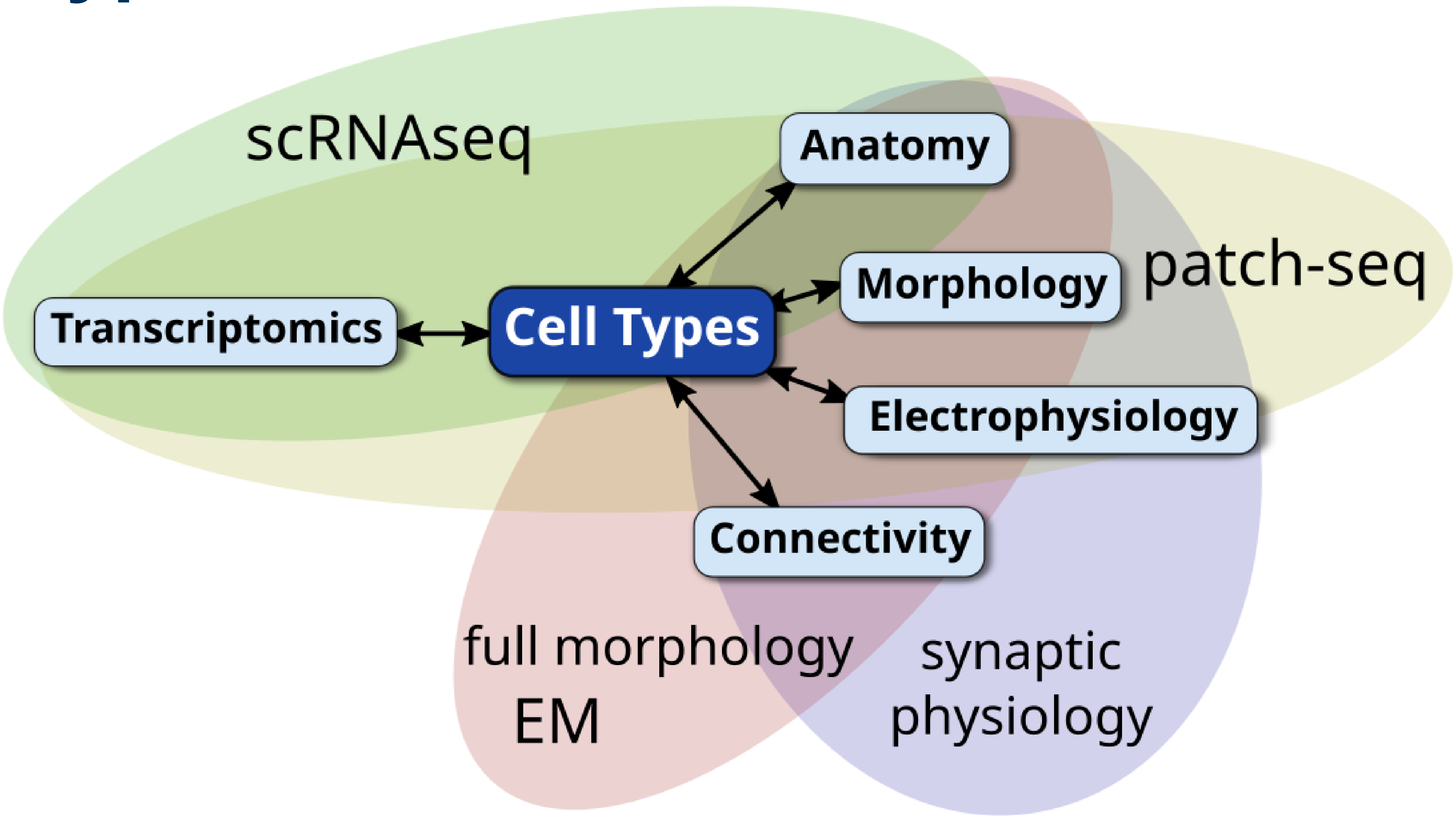
To aid ongoing efforts towards interneuron classification, to facilitate the exchange of information and to build a foundation for future progress in the field, we propose and publicly endorse a standardized nomenclature of interneuron properties. This proposal arose out of a meeting devoted to this topic in Cajal's native town, Petilla de Aragón (Navarra, Spain), and is rooted in the collective work that has been performed in many laboratories. We seek agreement on a list of the essential features that differentiate the GABAergic interneurons of the neocortex, and encourage all investigators to use the same terms to describe the same features. We hope that this first step will later lead to a broadly adopted classification. However, before we are able to speak a common language, we need to agree on the words.

PING (2008)

How we define cell types today at Allen

- **Transcriptomics: RNA Expression. Genome = DNA, transcriptome = RNA**
- Define cell types based on gene expression (transcriptomics)
  - High throughput (millions of cells)
  - Many features (~30,000+ genes)
  - Direct links to genome (disease/genetic tools)
  - Works with postmortem tissue
  - Universal across organs and species
  - Many analysis tools
- Map knowledge from other modalities onto these transcriptomics-based definitions

Matching other modalities to gene-based cell type taxonomies



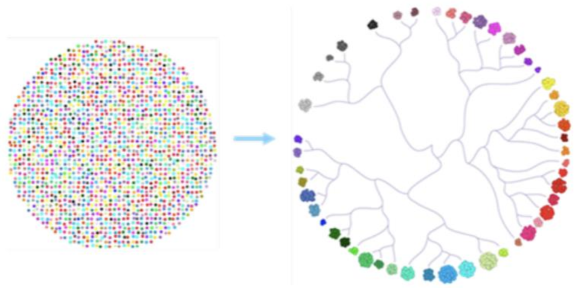
Still need a standard taxonomy format

- In June 2019 over 60 representatives from the fields of ontology, taxonomy, and neuroscience made recommendations, highlighted best practices, and **proposed conventions for naming cell types.**
- Participants included
    - Lindsay Cowell, UT Southwestern Medical Center
    - Sean Hill, University of Toronto
    - Deep Ganguli, CZI
    - Junhyong Kim, University of Pennsylvania
    - Evan Macosko, Broad
    - Maryann E. Martone, UCSD
    - Sean Mooney, University of Washington
    - David Osumi-Sutherland, EMBL-EBI
    - Rahul Satja, NY Genome Center

    …and many more

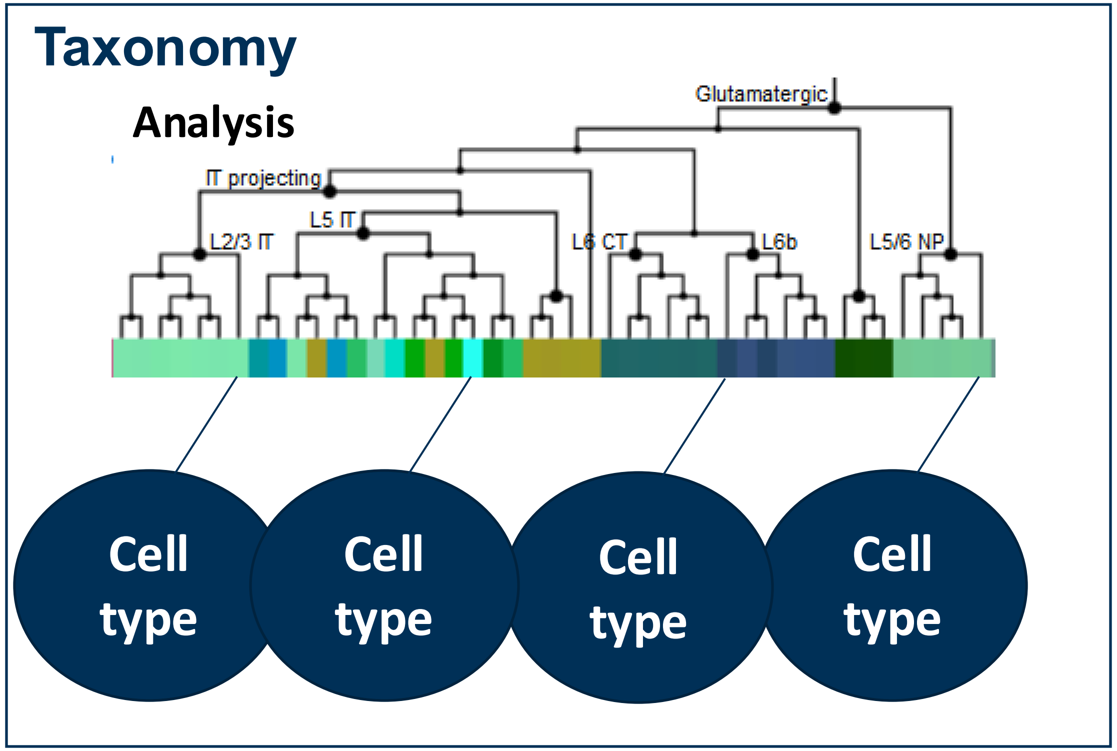
Defining an Ontological Framework for a Brain Cell Type Taxonomy: Single Cell -omics and Data-Driven Nomenclature

June 17-18, 2019
Allen Institute, Seattle, WA



Result: Common Cell Type Nomenclature (CCN)

1. Schema



Taxonomies must have unique IDs, aliases, and other meta-data to allow for tracking of information and alignment between studies.

2. Naming Convention (in mammalian cortex)

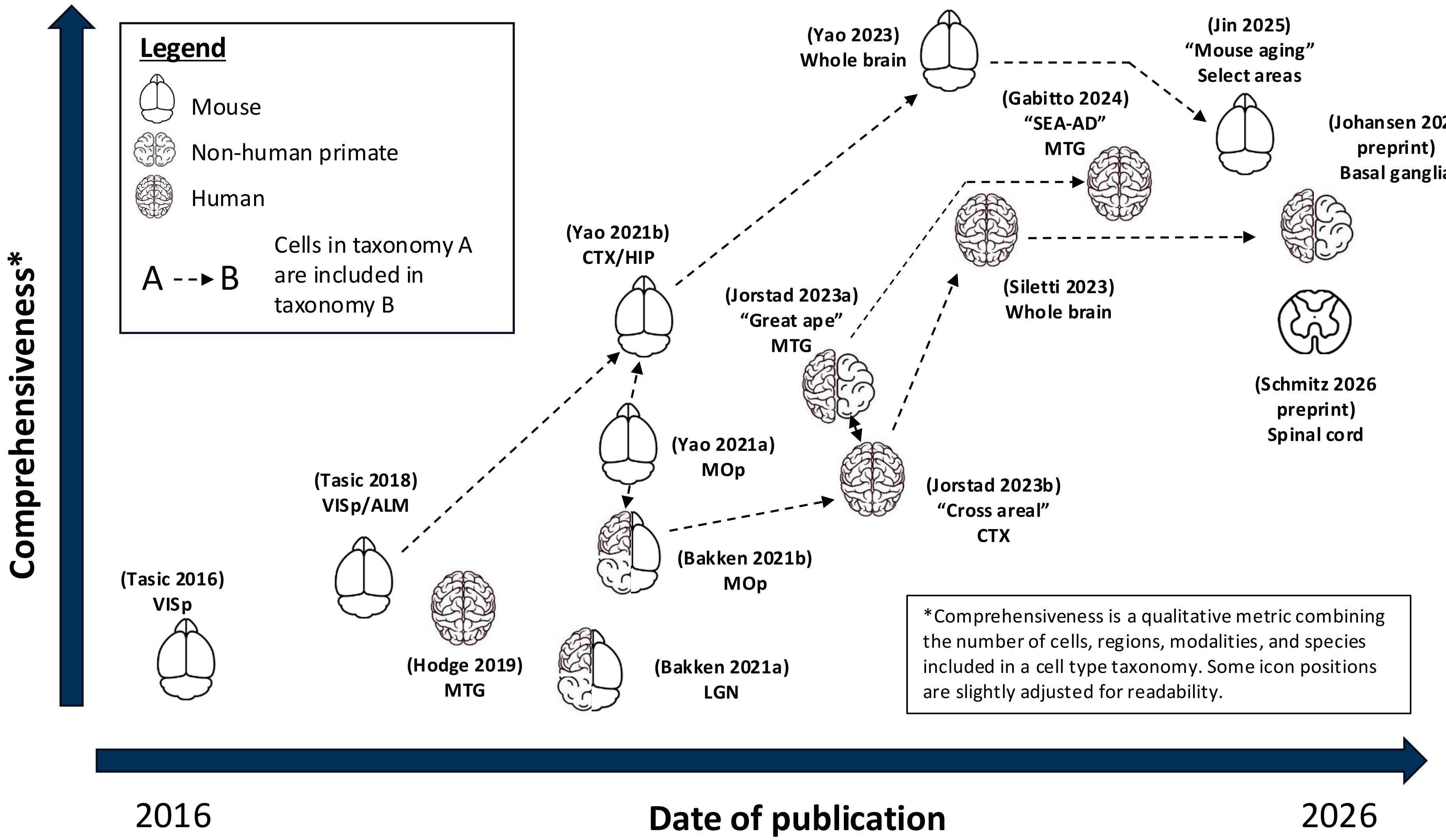
Mini-atlas to have aligned alias with alternative alias.

Aligned alias: Pvalb
Alternative alias: basket & chandelier

Class	Format	Example
Glutamatergic	[Layer] [Projection] #	L2/3 IT 4
GABAergic	[Canonical gene] #	Pvalb 3
Non-neuronal	[Cell class] #	Microglia 2
(Any)	[Historical name] #	Chandelier 1

<u>More details in Miller (2020)</u>

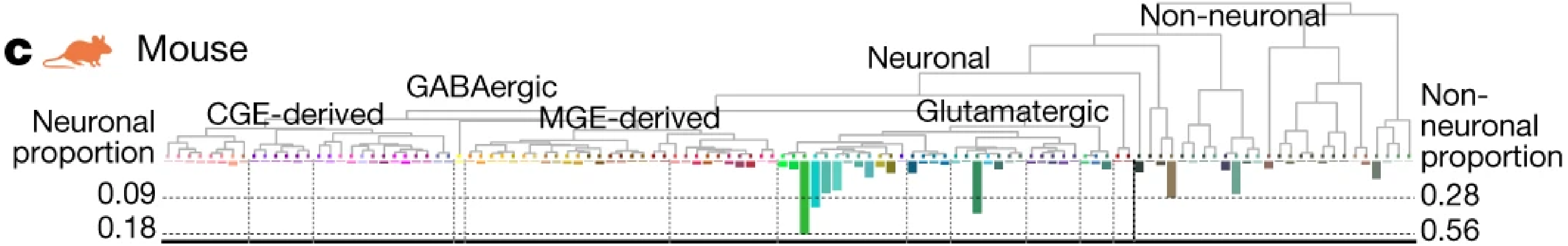
A history of taxonomies from the Allen Institute: 2016- 2026



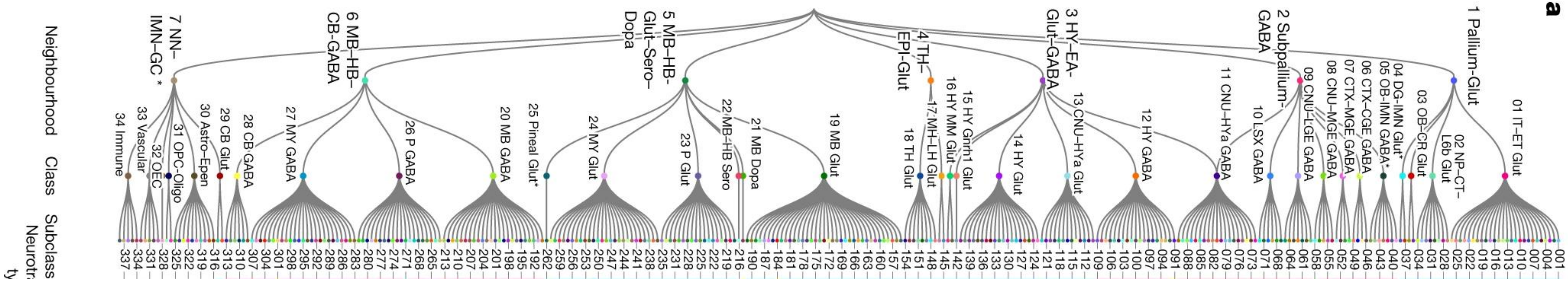
Credit: Jeremy Miller

How can we visualize taxonomies?

Mouse M1 – 116 cell types



Whole mouse brain – 5,322 cell type clusters



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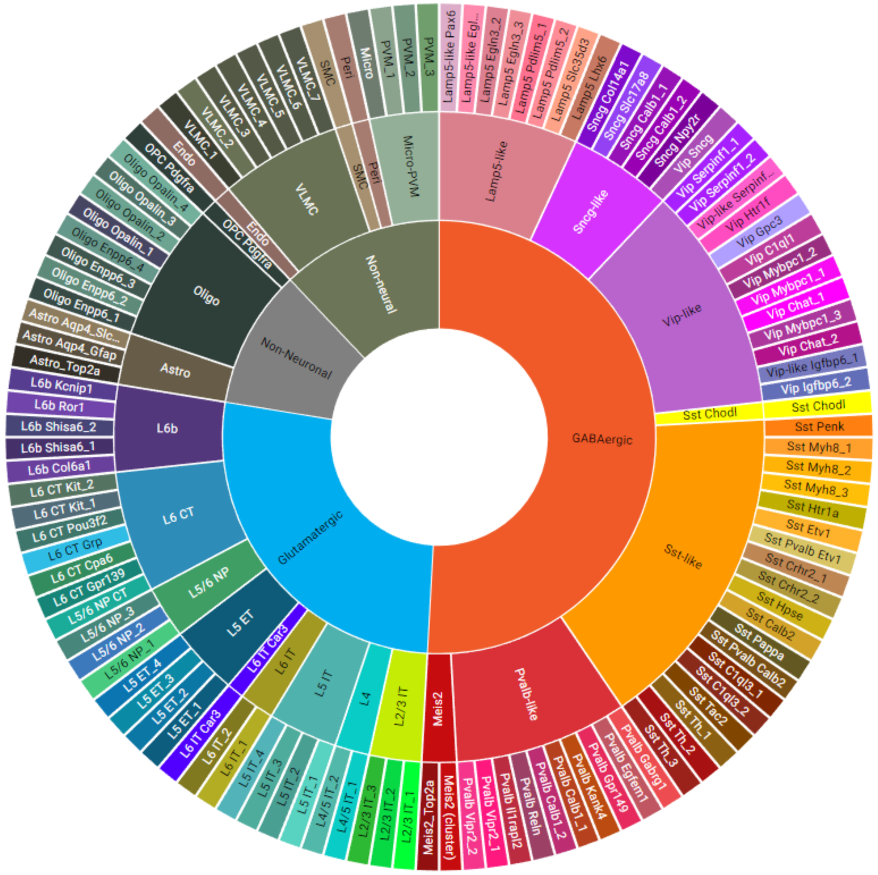

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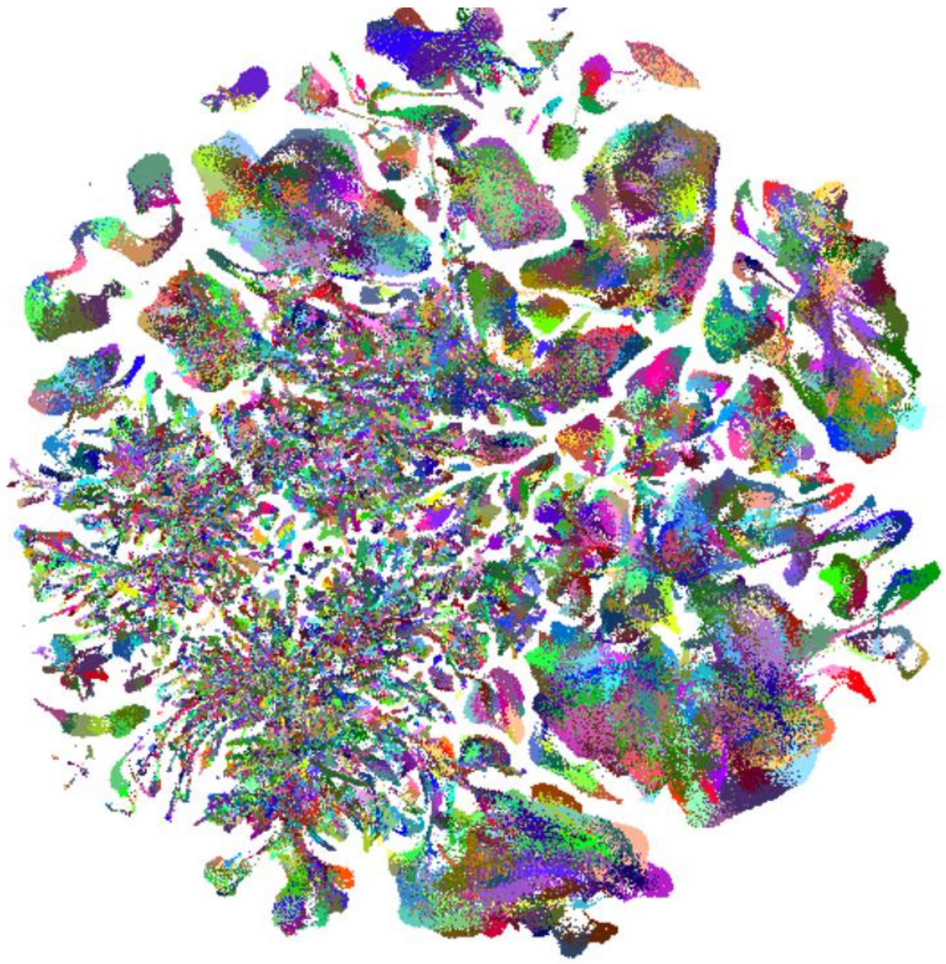

What to do with all these data??

Using Allen transcriptomic data for <u>transcriptomic</u> experiments – data annotation

Single-Nucleus Profiling Identifies Accelerated Oligodendrocyte Precursor Cell Senescence in a Mouse Model of Down Syndrome

Bianca Rusu,[1,2] Bharti Kukreja,[2] Taiyi Wu,[2] Sophie J. Dan,[2,3,4] Min Yi Feng,[1,2] and Brian T. Kalish[1,2,5]

https://doi.org/10.1523/ENEURO.0147-23.2023

differential expression. Major cell type annotations were assigned to clusters by manual inspection of canonical marker gene signals and confirmed through scPred's reference-based mapping approach (v1.9.2; Alquicira-Hernandez et al., 2019) using the Allen Brain Atlas whole cortex and hippocampus 10× dataset (Yao et al., 2021).

MapMyCells on Wednesday!

Using transcriptomic data for non-transcriptomic experiments – experimental design

Genetically Defined Subtypes of Somatostatin-Containing Cortical Interneurons

Rachel E. Hostetler, Hang Hu, and Ariel Agmon

https://doi.org/10.1523/ENEURO.0204-23.2023

To develop genetic tools for accessing distinct subtypes of SOM interneurons, we searched for available mouse driver lines in which Cre recombinase was coexpressed with marker genes for identified transcriptomic SOM groups (Tasic et al., 2018). We selected the following four such lines:

Using transcriptomic data for <u>non-transcriptomic</u> experiments – experimental design

Id2 GABAergic interneurons comprise a neglected fourth major group of cortical inhibitory cells

Robert Machold[1], Shlomo Dellal[1†], Manuel Valero[1†‡], Hector Zurita[1], Ilya Kruglikov[1], John Hongyu Meng[1,2], Jessica L Hanson[1§], Yoshiko Hashikawa[1], Benjamin Schuman[1], György Buzsáki[1,3], Bernardo Rudy[1,3,4*]

We examined cortical single-cell transcriptome data (scRNAseq) from the Allen Institute (Tasic et al., 2016) and observed that expression of the Id2 gene was highly enriched in the INs in their collection that do not express PV, SST, or VIP (Figure 1A; Mayer et al., 2018). The Id2 IN population

Questions?

Nomenclature Game

<u>Organize your cell types into categories, trees, whatever you think is right!</u>

After sorting, we will compare groups